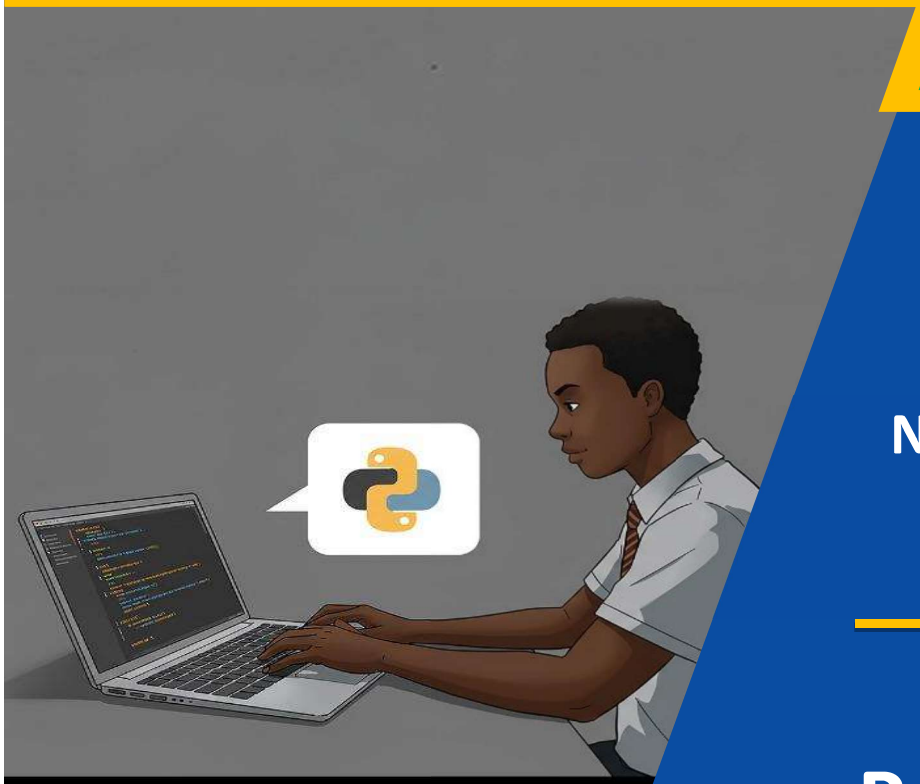
RQF LEVEL 5



GENPP501

NETWORKING AND INTERNET TECHNOLOGIES

Python Programming Fundamentals

TRAINEE'S MANUAL

October, 2024


cmd-python
Microsoft Windows [version 10.0.22621.3593]
(c) Microsoft Corporation. All rights reserved.
C:\Users\samsung>pip install nltk
Collecting nltk
Downloading nltk-3.8.1-py3-none-any.whl (1.5 MB)
Collecting click
Downloading click-3.8.1-py3-none-any.whl (97 kB)
Collecting joblib
Downloading joblib-1.4.2-py3-none-any.whl (301 kB)
Collecting nltk
Downloading nltk-3.8.1-py3-none-any.whl (97 kB)
Collecting regex>=2021.8.3
Downloading regex-2024.10.22-cp311-win_amd64.whl(268 kB)
Collecting colorama
Downloading colorama-0.4.6-py3-none-any.whl (25 kB)
Installing collected packages: regex, joblib, colorama, click, nltk
Successfully installed click-8.1.7 colorama-0.4.6 joblib-1.4.2 nltk-3.8.1 regex-2024.10.22
[notice] A new release of pip available : 22.3.1
[notice] To update, run: python.exe -m pip install --upgrade pip

Indicative content 1.1: Selection of Python Tools

Duration:5 hrs

Theoretical Activity 1.1.1: Description of python programming

Tasks:

1. You are requested to answer the following questions:
    i. Define python
    ii. Discuss on benefits of python
    iii. Explain characteristics of python
    iv. Python programming is applied in different areas, describe each of the following areas:
        a) Data science
        b) Software development
        c) Automation
        d) Data analysis
2. Write your findings on papers
3. Present your findings to the whole class or trainer
4. For more clarification read key reading 1.1.1 and ask questions where necessary.

Key readings 1.1.1.Description of python programming

1. Definition

Python is a high-level, interpreted programming language known for its simplicity and readability. Created by Guido van Rossum and first released in 1991, Python has gained immense popularity due to its versatility and wide range of applications.

It is used for:

- web development (server-side),
- software development,
- mathematics,
- System scripting.

2. Uses of Python

Python is used in the following ways:

- Python can be used on a server to create web applications.
- Python can be used alongside software to create workflows.
- Python can connect to database systems. It can also read and modify files.

- Python can be used to handle big data and perform complex mathematics.
- Python can be used for rapid prototyping, or for production-ready software development.

3. Comparison between Python Syntax and other programming languages

- Python was designed for readability, and has some similarities to the English language with influence from mathematics.
- Python uses new lines to complete a command, as opposed to other programming languages which often use semicolons or parentheses.
- Python relies on indentation, using whitespace, to define scope; such as the scope of loops, functions and classes. Other programming languages often use curly-brackets for this purpose.

4. Benefits of python

Python offers a range of advantages that make it a popular choice for developers, data scientists, and researchers.

4.1. Simplicity and Readability

Python's clear syntax and structure promote easy reading and writing of code, which is especially beneficial for beginners.

4.2. Versatility

Python is suitable for various applications, including web development, data analysis, artificial intelligence, scientific computing, automation, and more.

4.3. Extensive Libraries and Frameworks

Python boasts a rich ecosystem of libraries and frameworks (e.g., NumPy, pandas, Django, Flask) that simplify complex tasks and speed up development.

4.4. Strong Community Support

A large and active community means extensive support, documentation, and resources, making it easier for learners to find help and tutorials.

4.5. Cross-Platform Compatibility

Python can run on various operating systems (Windows, macOS, Linux), allowing for flexibility in development and deployment.

4.6. Dynamic Typing

Variables in Python do not require an explicit declaration of data types, which can speed up coding and reduce boilerplate.

4.7. Interpreted Language

Python code is executed line by line, which simplifies debugging and makes it easier to test code snippets.

4.8. Integration Capabilities

Python can easily integrate with other languages (like C, C++, and Java) and technologies, making it a good choice for projects that require mixed-language support.

4.9. Object-Oriented and Functional Programming

Python supports both programming paradigms, allowing developers to choose the most suitable approach for their projects.

4.10. Rapid Development

The combination of an easy learning curve, extensive libraries, and rapid prototyping capabilities allows for quicker development cycles.

5. Characteristics of python

Python is a high-level, general-purpose programming language known for its simplicity, readability, and versatility. Here are some of its key characteristics:

5.1. Readability

Clear syntax: Python uses indentation to define code blocks, making it easy to read and understand.

English-like keywords: Python uses words like "if", "else", "for", and "while" to control program flow, making it more intuitive for beginners.

5.2. Versatility

Wide range of applications: Python can be used for web development, data analysis, scientific computing, machine learning, automation, and more.

Cross-platform compatibility: Python code can run on various operating systems like Windows, macOS, and Linux.

5.3. Efficiency

- **Interpreted language:** Python code is executed line by line, making it slower than compiled languages like C++. However, it's often faster than other interpreted languages like Ruby.
- **Large standard library**: Python comes with a rich standard library that provides modules for various tasks, reducing development time.

5.4. Community and Ecosystem

- **Active community**: Python has a large and supportive community of developers, which means you can find plenty of resources, tutorials, and help online.
- **Extensive ecosystem**: Python has a vast ecosystem of third-party libraries and frameworks, such as NumPy, Pandas, TensorFlow, and Django, that extend its capabilities.

5.5. Other characteristics

- **Dynamic typing**: Python automatically determines the data type of variables at runtime.

- **Object-oriented programming**: Python supports object-oriented programming paradigms, allowing you to create modular and reusable code.
- **Memory management**: Python handles memory management automatically, freeing developers from the burden of manual memory allocation and deallocation.

6. Application of python

The applications of Python programming are diverse and span various fields.

6.1. Data Science

- **Data analysis:** Python's libraries like NumPy, Pandas, and Matplotlib facilitate data cleaning, exploration, and visualization.
- **Machine learning:** Frameworks like TensorFlow and PyTorch enable building and training complex machine learning models.
- **Deep learning:** Python is widely used for developing deep neural networks for tasks like image recognition and natural language processing.

6.2. Software Development

- **Web development:** Frameworks like Django and Flask streamline the creation of web applications.
- **Desktop applications:** Libraries like Tkinter, PyQt, and wxPython allow for building graphical user interfaces.
- **Game development:** Pygame is a popular framework for creating simple games.

6.3. Automation:

- **Task automation:** Python can automate repetitive tasks, such as sending emails, downloading files, or interacting with web applications.
- **System administration:** Python scripts can be used to automate system management tasks, like backing up data or monitoring system performance.

6.4. Data Analysis:

- **Statistical analysis:** Python's libraries like SciPy and Stats models provide tools for statistical analysis and modelling.
- **Data mining:** Python can be used for discovering patterns and trends in large datasets.
- **Data visualization**: Matplotlib and Seaborn allow for creating informative and visually appealing charts and graphs.



Practical Activity 1.1.2: Selecting python tools



Task:

1. Read the task bellow
   Ubumwe ltd need to develop a web app that help them in selling their products online and the system will have capabilities to perform automatic updates while products are sold and automatic deployment on the side of system administrator. You are hired as machine learning engineer responsible for selecting the best tools that will be used.
2. Referring to the provided key reading 1.1.2, perform the task described above.
3. Present your work to the trainer and whole class.



Key readings 1.1.2 Selection of python tools

1. Criteria for selecting tools

Selecting the right Python tools for your project can significantly impact your productivity and the quality of your work.

1.1. Purpose and Requirements

Define the primary goal of your project. Are you building a web application, performing data analysis, or developing a machine learning model?

1.2. Ease of Use

Consider how user-friendly the tool is. Does it have a vertical learning curve, or can you get started quickly? Tools with good documentation and community support can be very helpful.

1.3. Integration

Check if the tool integrates well with your existing systems and workflows. For example, if you're using a specific database or framework, ensure the tool supports it.

1.4. Scalability

Evaluate whether the tool can handle the scale of your project. If you're working with large datasets or high-traffic applications, you'll need tools that can scale accordingly.

1.5. Performance

Assess the performance of the tool. Some tools may be faster or more efficient than others, which can be crucial for performance-critical applications.

1.6. Community and Support

A strong community and good support can make a big difference. Look for tools with active communities, frequent updates, and responsive support.

1.7. Security

Ensure the tool has robust security features, especially if you're dealing with sensitive data.

1.8. Cost

Consider the cost of the tool. While some tools are free and open-source, others may require a subscription or one-time purchase.

1.9. Flexibility

The tool should be flexible enough to adapt to your project's evolving needs. It should allow for customization and extension.

1.10. Compatibility

Ensure the tool is compatible with your operating system and other tools you plan to use.

N.B: Among the tools you have to select the followings depending on the project to be developed.

2. Python IDE

An IDE is a software application that provides comprehensive facilities for programmers to develop software.

Here are some of the most popular Python IDEs:

2.1. Visual Studio Code (VS Code)

Platform

Windows, macOS, Linux

Features

Lightweight, highly customizable, supports various programming languages, intelligent code completion, debugging, Git integration, and a vast extension marketplace.

2.2. PyCharm

Platform: Windows, macOS, Linux

Features: Specifically designed for Python development, intelligent code completion, refactoring, debugging, unit testing, and integration with popular Python frameworks like Django and Flask.

2.3. Jupiter Notebook

Platform: Web-based, can be used in most browsers

Features: Interactive environment for data analysis, scientific computing, and machine learning. It's great for exploring data, creating visualizations, and sharing results.

2.4. Spyder

Platform: Windows, macOS, Linux

Features: Similar to MATLAB, Spyder is a scientific computing environment with a MATLAB-like interface, suitable for data analysis and scientific programming.

2.5. Sublime Text

Platform: Windows, macOS, Linux

Features: A lightweight, highly customizable text editor with excellent Python support, including syntax highlighting, code completion, and multiple panes.

2.6. Thonny

Platform: Windows, macOS, Linux

Features: A beginner-friendly IDE designed for teaching programming, with a simple interface, step-by-step debugging, and a built-in Python interpreter.

3. Python frameworks

Python frameworks provide a pre-built structure and tools to streamline the development process. They offer reusable code, best practices, and a modular approach, making it easier to build applications efficiently.

3.1. Web Development Frameworks

3.1.1. Django:

A full-stack framework known for its rapid development, scalability, and security features. It's ideal for complex web applications.

3.1.2. Flask

A lightweight, minimalist framework that offers flexibility and control. It's suitable for smaller projects and APIs.

3.1.3. FastAPI

A modern framework that emphasizes performance, ease of use, and developer experience. It's built on top of Starlette and Pedantic.

3.2. Data Science and Machine Learning Frameworks

3.2.1. TensorFlow

A popular open-source platform for machine learning, deep learning, and natural language processing. It's used for building and training neural networks.

3.2.2. PyTorch

Another popular framework for machine learning, known for its flexibility and ease of use. It's often used for research and prototyping.

3.2.3. Scikit-learn

A machine learning library that provides a simple interface for building and training models, including classification, regression, clustering, and dimensionality reduction.

3.3. Scientific Computing and Data Analysis Frameworks

3.3.1. NumPy

A fundamental package for numerical computing, providing multi-dimensional arrays and matrices, along with mathematical functions.

3.3.2. Pandas

A powerful data analysis library offering data structures like Data Frames and Series, which make it easy to manipulate and analyse data.

3.3.3. Matplotlib

A plotting library for creating static, animated, and interactive visualizations.

Other Notable Frameworks:

3.3.4. Kivy

A framework for building cross-platform mobile apps, desktop applications, and web applications using a single codebase.

3.3.5. Twisted

A framework for asynchronous networking, making it suitable for building network-intensive applications like servers and chat clients.

3.3.6. CherryPy

A minimalistic web framework that emphasizes simplicity and performance.

3.4. Automation libraries in python

Python has several powerful libraries for automation that can help streamline tasks across various domains.

Here are some of the most popular ones:

3.4.1. Selenium

Use Case: Automating web browsers.

Description: Selenium allows you to control a web browser programmatically. It can be used for testing web applications and scraping data from websites.

3.4.2. PyAutoGUI

Use Case: GUI automation.

Description: PyAutoGUI lets you simulate mouse movements, clicks, and keyboard inputs to automate interactions with desktop applications.

3.4.3. requests

Use Case: HTTP requests.

Description: While primarily used for making HTTP requests, it can be leveraged for automating interactions with web APIs.

3.4.4. BeautifulSoup

Use Case: Web scraping.

Description: BeautifulSoup is used for parsing HTML and XML documents. It helps in extracting data from web pages.

3.4.5. Pandas

Use Case: Data manipulation and analysis.

Description: Pandas is excellent for automating data processing tasks, such as cleaning, transforming, and analyzing data in tabular formats.

3.4.6. Airflow

Use Case: Workflow automation.

Description: Apache Airflow is a platform to programmatically author, schedule, and monitor workflows, making it ideal for batch data processing.

3.4.7. Celery

Use Case: Distributed task queue.

Description: Celery is used for handling asynchronous tasks and scheduling them, making it great for background job processing.

3.4.8. Paramiko

Use Case: SSH and SFTP.

Description: Paramiko allows you to automate SSH connections and file transfers, useful for server management and automation.

3.4.9. Fabric

Use Case: SSH command execution.

Description: Fabric is a high-level Python library for executing shell commands remotely over SSH, making it easier to deploy applications.

3.4.10. pywinauto

Use Case: Windows GUI automation.

Description: This library allows you to automate GUI interactions on Windows applications.

3.4.11. Schedule

Use Case: Job scheduling.

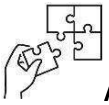
Description: A simple library for scheduling Python functions to run at specific intervals.



Points to Remember

- Python is a versatile programming language known for its simplicity and readability.
- Python has several benefits including: Simplicity and Readability, Versatility, Extensive Libraries and Frameworks, Strong Community Support, Cross-Platform, Compatibility, Dynamic Typing, Interpreted Language, Integration Capabilities, Object-Oriented and Functional Programming, Rapid Development.
- Key characteristics of python include: Readability, Versatility, Efficiency and Community and Ecosystem.

- Python can be used in data science and machine learning by using the following frameworks: TensorFlow, PyTorch and Scikit-learn.
- Scientific Computing and Data Analysis Frameworks: NumPy, Pandas, Matplotlib.
- In python we can use the following automation Libraries: Selenium, PyAutoGUI, Requests, Beautiful Soup, Airflow, Celery, Paramiko, Fabric, Pywinauto, And Schedule.
- Selecting the right Python tools for your project is crucial for productivity and quality. Start by defining your project's primary goal, whether it's web development, data analysis, or machine learning.
- Consider the tool's ease of use, integration with existing systems, scalability, and performance. A strong community and support, robust security features, and cost are also important factors.
- Ensure the tool is flexible, customizable, and compatible with your operating system and other tools you plan to use.
- Python support the following IDEs: VS Code, PyCharm, Jupyter Notebook, Spyder, Sublime Text and Thonny
- Python support the following web frameworks: Django, Flask and FastAPI
- Python can be used in data science and machine learning by using the following frameworks: TensorFlow, PyTorch and Scikit-learn
- Scientific Computing and Data Analysis Frameworks: NumPy, Pandas and Matplotlib
- In python we can use the following automation Libraries: Selenium, PyAutoGUI, requests, BeautifulSoup, Airflow, Celery, Paramiko, Fabric, Pywinauto and Schedule.



Application of learning 1.1.

HH ltd want to develop a system that will be used while selling their products online and performing some automations once new features have added to their website and performing automatic deployment. You are hired as machine learning engineer responsible for selecting the right tools that will be used to develop that software.

Indicative content 1.2: Installation of Python Tools

Duration: 5 hrs

Theoretical Activity 1.2.1: Identification of computer system requirements

Tasks:

1. You are requested to identify the system requirements to install python tools.
2. Write your findings on paper/flipchart
3. Present your findings to the whole class and trainer
4. For more clarification read key reading 1.2.1 and ask questions where necessary.

Key readings 1.2.1.: Identification of Computer System Requirements to Install Python Tools

The following are requirements to install Python tools in Computer System:

1. Hardware Requirements

Processor: A modern processor (Intel or AMD) with at least dual-core is recommended for running Python tools efficiently.

For data science, machine learning, or automation tasks, a multi-core processor or GPU (for TensorFlow or PyTorch) can significantly enhance performance.

Memory (RAM): A minimum of 4 GB of RAM is recommended for general Python development.

For more resource-intensive tasks like data analytics, machine learning, or web development, 8 GB to 16 GB of RAM is ideal.

Storage: At least 1 GB of free disk space is needed to install Python and related tools.

For larger projects or when working with large datasets, SSD storage is preferred for faster read/write speeds.

Graphics Card: If working with AI/ML libraries (e.g., TensorFlow, PyTorch), a dedicated NVIDIA GPU with CUDA support is beneficial for model training.

2. **Software Requirements**

Operating System: Python tools can run on Windows, macOS, or Linux. However, certain tools may have better support on Linux and macOS (e.g., TensorFlow on GPUs). Ensure that the OS version is up-to-date and compatible with Python versions (e.g., Windows 10 or higher).

Python Interpreter: The latest version of Python (3.x) should be installed. Some tools may require specific versions of Python (3.6, 3.8, etc.).
Package Manager: pip (Python's package installer) is required for installing most Python tools. Ensure that pip is installed and up-to-date.
Development Tools: A suitable IDE or text editor (such as PyCharm, VS Code, or Jupyter Notebook) is required for writing and testing Python code.
If working with web development or DevOps, Node.js, Docker, or Git may be necessary for additional setup.



Practical Activity 1.2.2: Installing python software tools



Task:

1. Read the task bellow
   As machine learning engineer, you are asked to go to the computer lab to install python and PyCharm in a computer.
2. Referring to the provided key reading 1.2.2, perform the task described above.
3. Present your work to the trainer and whole class.



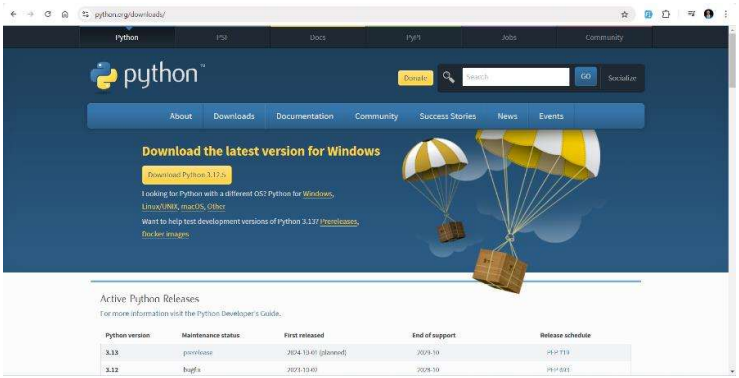
Key readings 1.2.2 Installation of python software tools

Install python software tools

1. Install python

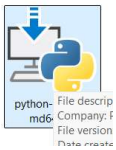
a. Download the Python Installer or use offline from external storage

- ✓ Go to the official Python website: https://www.python.org/downloads/
- ✓ Select the latest stable Python version (e.g., Python 3.12).
- ✓ Click on the appropriate installer for your Windows system (32-bit or 64-bit).



b. RUN the Installer

✓ Double-click the installer file (e.g., python-3.12.exe).



✓ Follow the on-screen instructions: Check the "Add Python to PATH" option to make Python accessible from the command line.

✓ Choose the installation location (default is usually fine).

✓ Click "Install Now" or "Customize installation" for more advanced options.

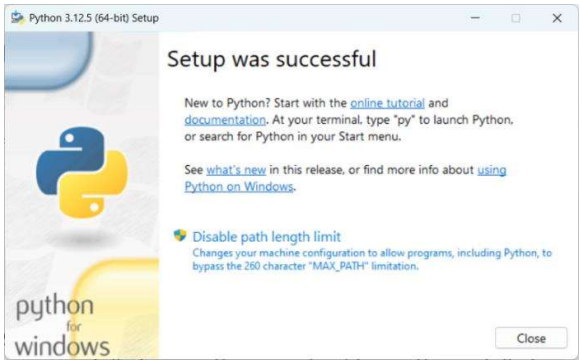


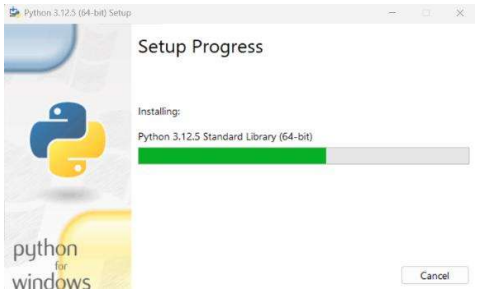
✓ Wait for the installation to complete.

C. Verify the Installation

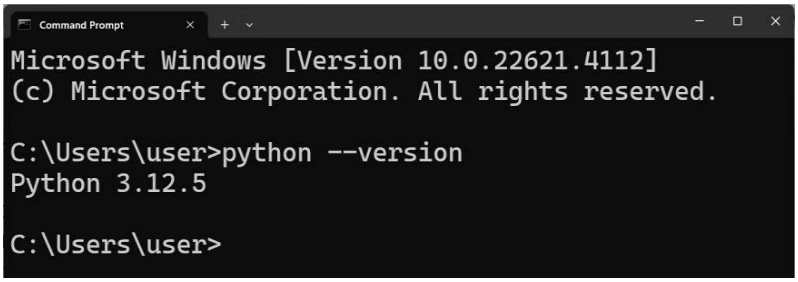
✓ Open a command prompt or terminal window.

✓ Type python --version and press Enter.





✓ If Python is installed correctly, you should see the installed version number displayed.

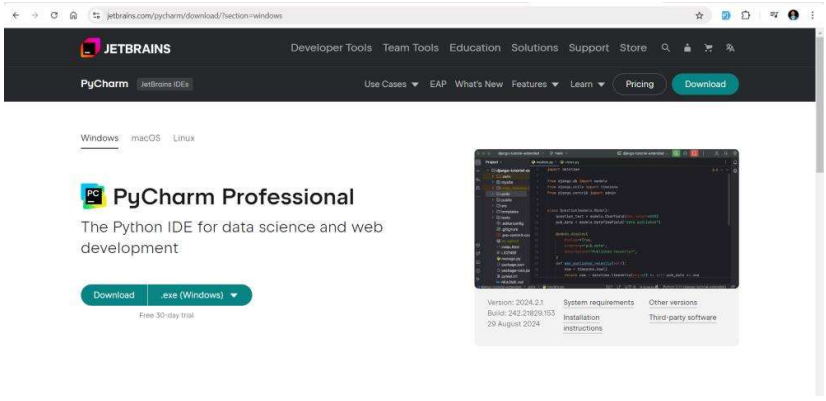


2. Install PyCharm

a. **Download PyCharm or use offline from external storage**

✓ Visit the Jet Brains website: https://www.jetbrains.com/pycharm/download/

✓ Click on the "Download PyCharm" button.



✓ Choose the appropriate edition for your needs (Community or Professional).

✓ Select the Windows installer.



b. **Run the Installer**

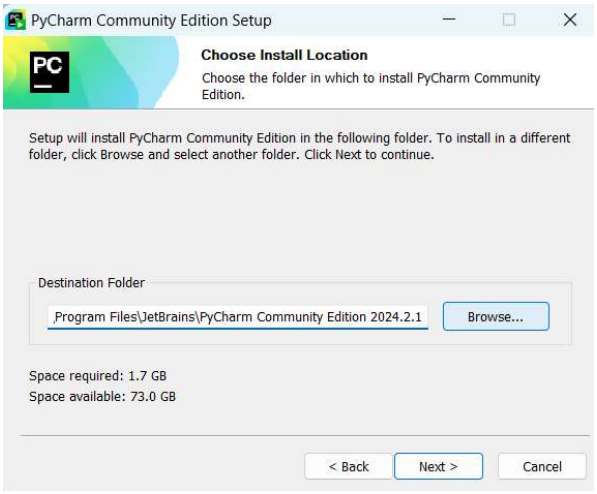
✓ Double-click the downloaded installer file (e.g., pycharm-community-2023.3.exe).



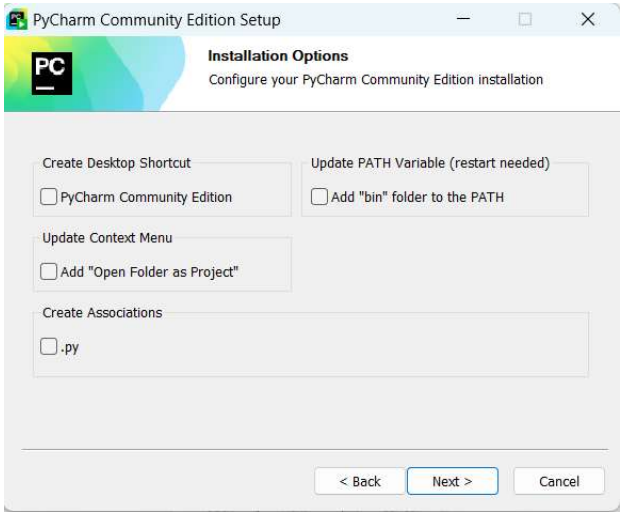
Follow the on-screen instructions

✓ Accept the license agreement.

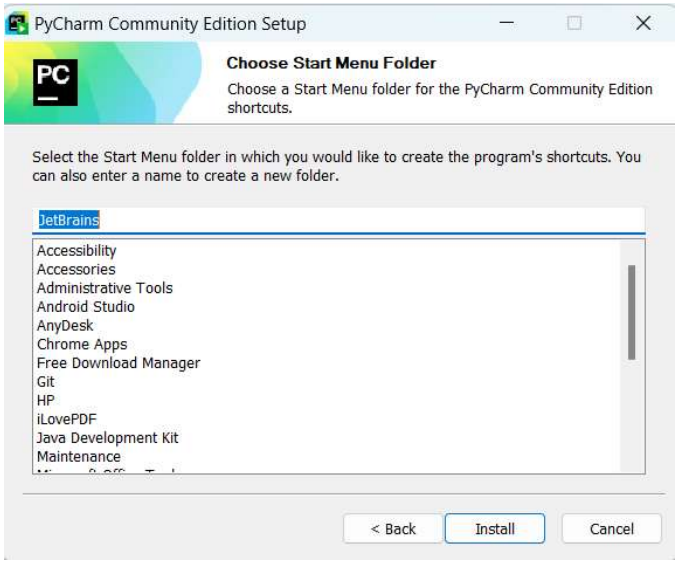
✓ Choose the installation location (default is usually fine).

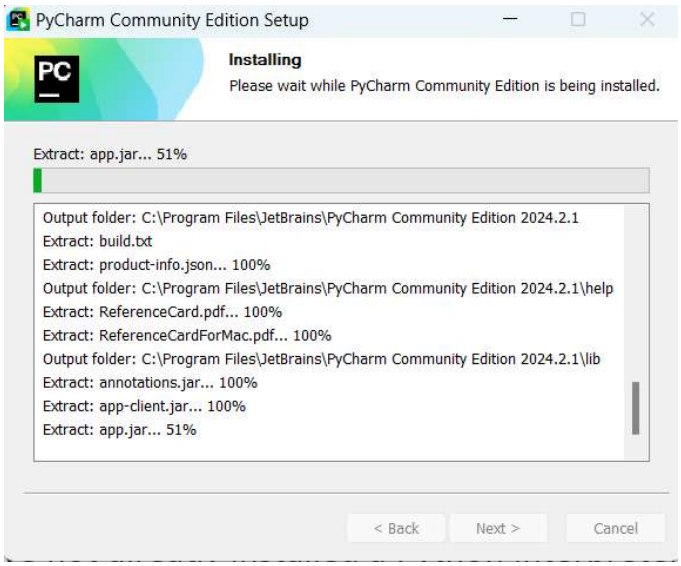


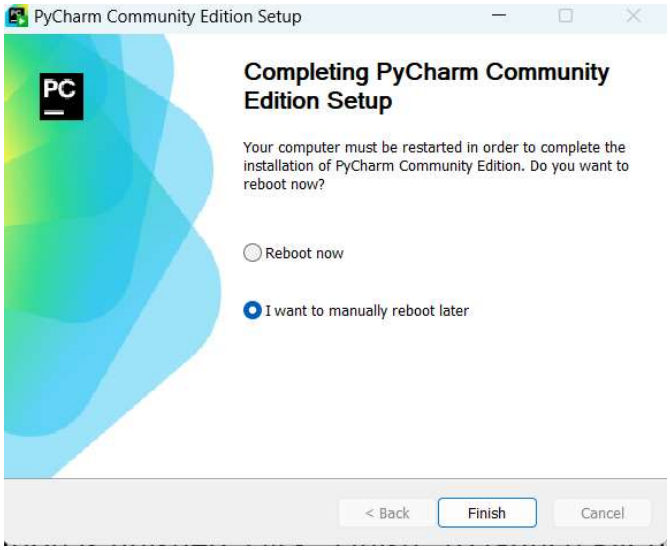
✓ Select the desired components to install (e.g., Python interpreter, web development tools).



✓ Click "Install" and wait for the installation to complete.

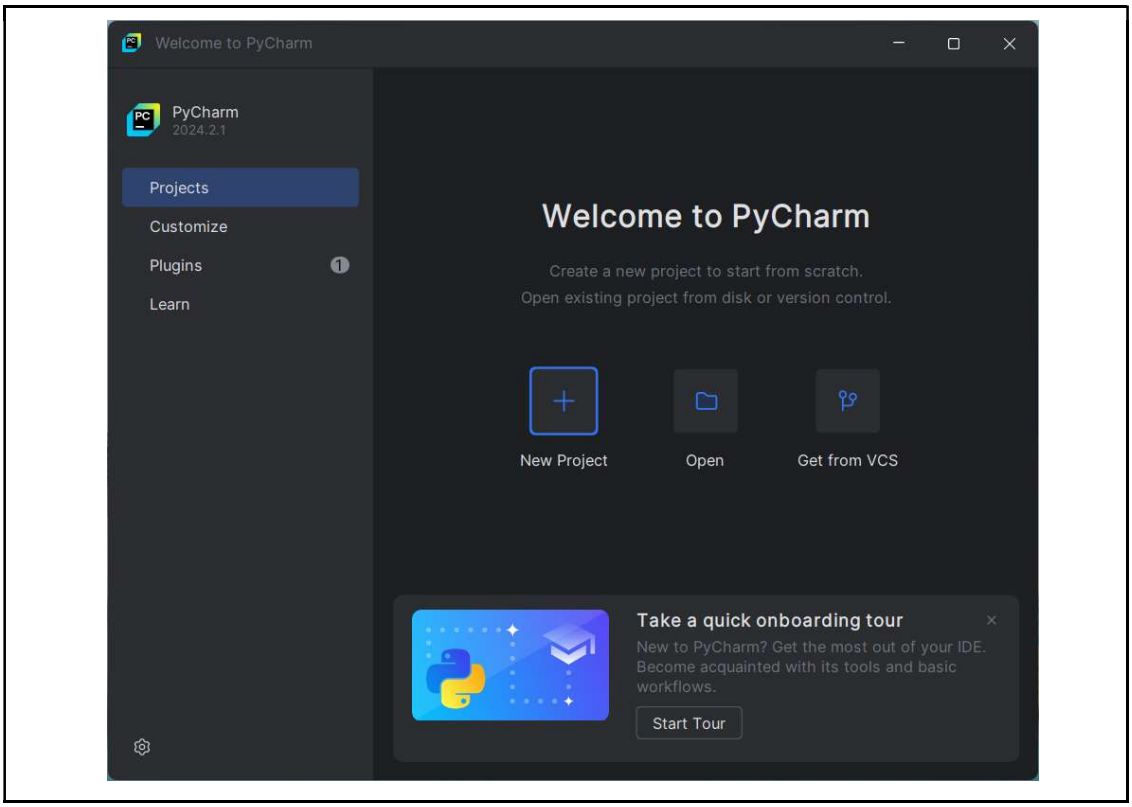






c. **Launch PyCharm**

Once the installation is finished, click "Finish" to launch PyCharm.



Practical Activity 1.2.3: Configuring python virtual environment

Task:

1. Read key reading 1.2.3. **Configuration of python virtual environment**
2. Referring to the Key reading 1.2.3 you are requested to go to the computer lab to configure python virtual environment.
3. Apply safety precautions.
4. Referring to the steps provided in key readings, configure python virtual environment.
5. Present out the steps to configure python virtual environment.

Key readings 1.2.3 Configuration of python virtual environment

1. Introduction

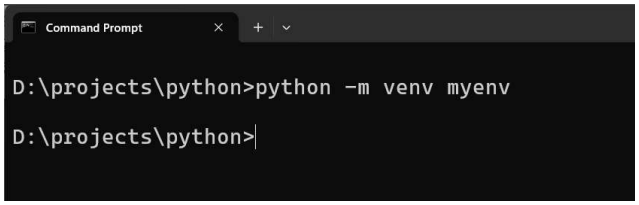
A Python virtual environment is a tool that helps you create an isolated environment for your Python projects. This means you can manage dependencies for each project separately, avoiding conflicts between different projects' requirements.

2. Creating a Virtual Environment

✓ Open your terminal or command prompt.

✓ Navigate to your project directory.

✓ Run the following command "**python -m venv myenv"**



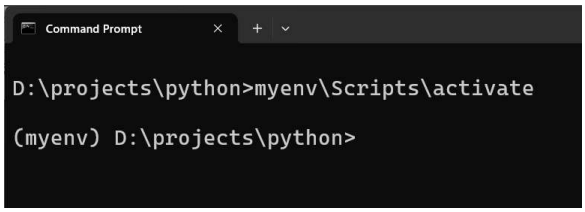
✓ This will create a directory named myenv containing a standalone Python installation.



3. Activating the Virtual Environment

✓ **On Windows**

Run the following command **myenv\Scripts\activate**



✓ **On macOS and Linux**

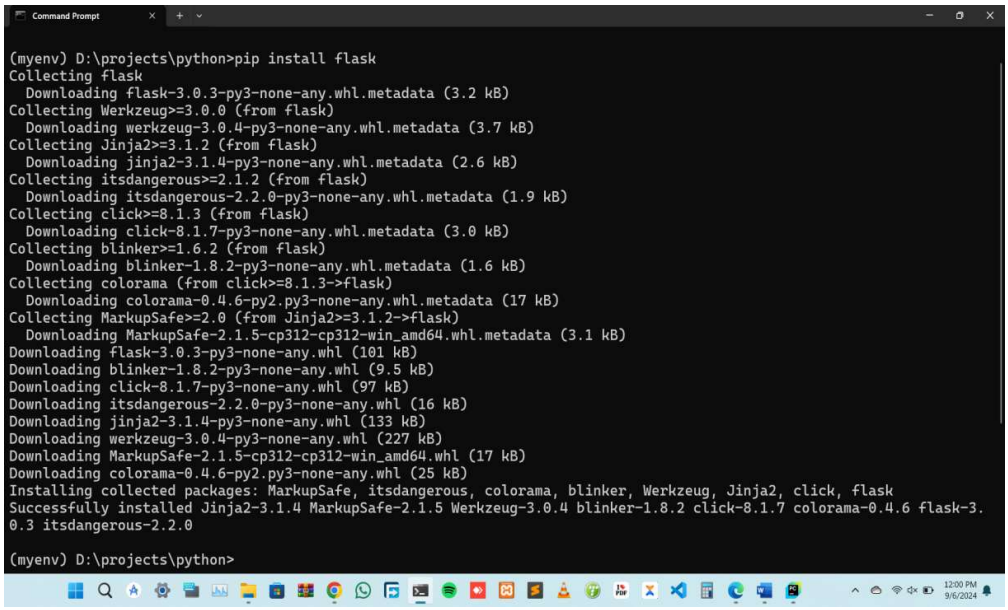
Run the following command **source myenv/bin/activate**

Once activated, your terminal prompt will change to indicate that you are now working within the virtual environment.

3. Installing Packages

With the virtual environment activated, you can install packages using pip

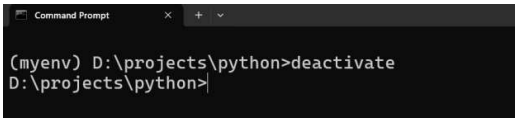
"pip install package name"



4. Deactivating the Virtual Environment

To exit the virtual environment, simply run

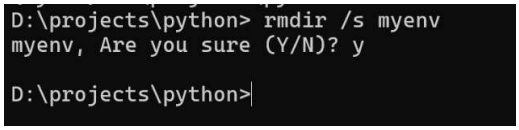
"deactivate"



5. Deleting the Virtual Environment

If you no longer need the virtual environment, you can delete the myenv directory

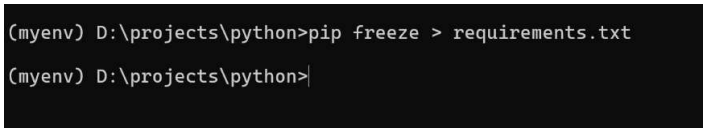
rmdir /s myenv



6. Freeze packages

The "freeze" command in the context of Python virtual environments is used to create a list of all installed packages and their versions. This is particularly useful for replicating your environment or sharing your project setup with others.

- ✓ Activate your virtual environment if it's not already activated.
- ✓ Run the freeze command



This command will create a file named requirements.txt in your current directory, containing a list of all installed packages and their versions.

7. To install packages from a requirements.txt file in another environment

This is helpful for setting up identical environments across different machines or for other developers working on the same project

```
(myenv) D:\projects\python>pip install -r requirements.txt
```



Points to Remember

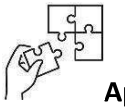
- To install Python tools, your system should meet certain hardware requirements: at least a dual-core processor, 4 GB RAM (preferably 8-16 GB for heavier tasks), 1 GB of free disk space, and optionally, a GPU for machine learning.
- On the software side, ensure you have the latest version of Python 3.x, an appropriate IDE (e.g., VS Code, PyCharm), pip for package management, and necessary dependencies like C++ build tools or Java for specific libraries
- While installing python we follow these steps:

  Step 1: download the Python Installer or use offline from external storage
  Step 2: run the installer
  Step 3: verify the installation

- while installing PyCharm we follow these steps:

Step 1: Download PyCharm or use offline from external storage
Step 2: Run the installer
Step 3: Launch PyCharm

- Python virtual environments create isolated environments for projects, preventing dependency conflicts.
- To create a virtual environment, use python -m venv myenv.
- Activate it using myenv\Scripts\activate.
- Install packages with pip install package_name.
- Deactivate using deactivate.
- Freeze dependencies with pip freeze > requirements.txt.
- Install from requirements.txt using pip install -r requirements.txt.



Application of learning 1.2.

HHT LTD is software development Company located in Kicukiro district, they want to develop a web app that will help in selling their products online and the system will have capabilities

to perform automatic deployment on the side of system administrator. You have been hired as machine learning engineer responsible for installing and configuring all python tools that will be needed to develop that project.
The company will provide all tools, materials and equipment.

Indicative content 1.3: Testing Python Installation

Duration: 5 hrs

Practical Activity 1.3.1: Testing python installation

Task:

1. Read key reading 1.3.1.

2: As machine learning engineer, you are asked to go to the computer lab to Run python version command, Check python interpreter and test package manager.

3. Apply safety precautions

4. Referring to the steps provided in key readings, Run python version command, Check python interpreter and Test package manager.

5. Present out the steps to Run python version command, Check python interpreter and Test package manager.

Key readings 1.3.1 Testing python installation

To test your Python installation, follow these steps:

Step 1: Check Python Version

- ✓ Open a Terminal or Command Prompt:
- ✓ Windows: Press Win + R, type cmd, and press Enter.
- ✓ macOS: Open Finder, go to Applications, then Utilities, and double-click on Terminal.
- ✓ Linux: Open your terminal application.
- ✓ Run the Python Version Command **"python –version"**
- ✓ or, for systems where Python 3 is installed as python3 **"python3 –version"**

Expected Output

```
C:\Users\jeand>python --version
Python 3.7.4

C:\Users\jeand>
```

Step 2: Check Python Interpreter

Open Python Interpreter

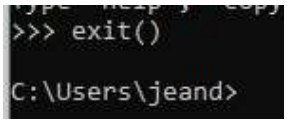
In the terminal, type "**python"** or **"python3"**

Verify the Prompt

```
C:\Users\jeand>python
Python 3.7.4 (tags/v3.7.4:e09359112e, Jul  8 2019, 20:34:20) [MSC v.1916 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

Exit the Interpreter

To exit, type **"exit()"** or press Ctrl + D (macOS/Linux) or Ctrl + Z followed by Enter (Windows).



Step 3: Test the Package Manager (pip)

Check pip Version

In the terminal, type **"pip –version"** or, for systems where pip is installed as pip3 **"pip3 –version"**

Expected Output

```
C:\Users\jeand>pip --version
pip 19.0.3 from c:\users\jeand\appdata\local\programs\python\python37\lib\site-packages\pip (python 3.7)
```

Test Installing a Package

As a test, you can try installing a simple package, such as Django, openpyxl, pytz:

pip install Django // for installing django

pip install openpyxl // for installing openpyxl

pip install pytz // for installing the time zone

Expected Output

You should see messages indicating that the package is being downloaded and installed.

```
Installing collected packages: et-xmlfile, openpyxl
Successfully installed et-xmlfile-1.1.0 openpyxl-3.1.3
You are using pip version 19.0.3, however version 24.0 is available.
You should consider upgrading via the 'python -m pip install --upgrade pip' command.

C:\Users\jeand>pip list
```

Verify the Package Installation

After installation, you can verify it by running **"pip list"**

This will display information about the packages that are installed, confirming it is installed.

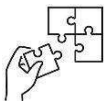
```
C:\Users\jeand>pip list
Package           Version
----------------- -------
asgiref           3.7.2
Django            3.2.25
et-xmlfile        1.1.0
openpyxl          3.1.3
pip               19.0.3
pytz              2024.1
setuptools        40.8.0
sqlparse          0.4.4
typing-extensions 4.7.1
You are using pip version 19.0.3, however version 24.0 is available.
You should consider upgrading via the 'python -m pip install --upgrade pip' command.

C:\Users\jeand>
```



Points to Remember

- To Check Python Version run that command in terminal: **python --version or python3 --version**
- To Check Python Interpreter run that command in terminal: **python or python3**
- To Test Package Manager: Check pip version: **pip –version** and Install a package: **pip install Django**
- To Verify installation run: **pip list**



Application of learning 1.3.

HHT LTD is software development company located in Kicukiro district, that company develop different software of institutions, they want to develop a web app that will help STY ltd in selling their products online and the system will have capabilities to perform automatic deployment on the side of system admistrator. You have been hired as machine learning engineer responsible for testing the python installation and installing required packages that will be used.

The company will provide all tools, materials and equipment.